Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/24/2005
PI Name:	Pyle, Barry H Ph.D.		
Project Title:	Bacterial Physiology and Virulence on Earth and in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	09/01/1999	**End Date:**	04/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Biorack		
Flight Assignment:	STS-107		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC2-1143		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project was based around an experiment that was planned to determine the effects of spaceflight and microgravity on bacterial growth, physiology, and virulence. Techniques used included biochemical and molecular techniques, virulence assays and physiological assays. A strain of Pseudomonas aeruginosa that produces exotoxin A was cultured on orbit and on earth to determine microgravity effects. Techniques that had been used on previous flight experiments such as plate counting, total direct microscopic counts, and metabolic fluorochromes were applied to assess growth and physiology. In addition, microscopic techniques including image analysis were employed to observe morphology, staining, and total cell count. Following the Columbia Shuttle disaster, efforts are being made to re-fly the experiment on a future Shuttle mission. To develop and perform a spaceflight experiment to determine the effects of microgravity on the physiology and virulence of Pseudomonas aeruginosa bacteria. Pseudomonas aeruginosa is a ubiquitous opportunistic bacterial pathogen that causes infections in people with weakened immunity. It has been found in		

	spacecraft water systems. Spaceflight crews may have secondary immunodeficiency, making them more susceptible to these infections particularly on long-duration missions like those proposed to the Moon and Mars, and extended International Space Station expeditions. In spaceflight conditions, bacteria may grow faster and reach higher population numbers, as well as becoming resistant to some antibiotics. Earth experiments with modeled microgravity suggest that bacterial virulence may also be enhanced. The combination of these factors suggests a greater potential health risk to spaceflight crew members. This research will provide evidence on whether or not spaceflight affects the physiology and virulence of this particular pathogen.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Exotoxin-producing Pseudomonas aeruginosa bacteria commonly cause hospital infections. Lowered immunity of space craft crews and earth bound individuals, and changes in bacterial characteristics in space flight indicate that a better understanding of Pseudomonas physiology and virulence is needed, both on earth and in space. Combining growth, physiology and virulence studies on earth and in spaceflight provides a unique opportunity to link effects of gravity and microgravity. Proteomic analysis will indicate genes that are activated or inactivated in this organism on earth, and hence indicate targets for further study. Ground experiments, in addition to the flight experiment, will extend our understanding of the physiology and virulence of this bacterial pathogen.
Task Progress:	Ground experiments using clinostat Modeled Microgravity (MMG) systems show little effect on production of the Exotoxin A virulence protein. Proteomic analysis identified 20 proteins that were up-regulated in response to MMG. Of these, 5 were identified as putative virulence-related proteins. Studies through the growth cycle of these bacteria in MMG suggested that there may be some effects on ETA production during the late logarithmic (exponential) phase and early stationary phase of growth. Two Science Verification Tests were performed to verify satisfactory performance of the spaceflight experiment according to revised flight scenarios. In both of these, a water-adapted culture was used as the inoculum, rather than freshly-cultured washed bacteria suspended in water. The water-adapted cultures are stable at ambient temperatures, whereas the washed suspensions are more stable at refrigeration temperatures. Lyophilized or frozen inocula did not perform as well as the fresh-washed suspensions or water-adpated cultures. These experiments showed that stowage onboard at 5 degC following incubation, during descent, and until the experiment containers are returned to the laboratory is preferable. Ambient temperatures during descent would be satisfactory for most of the proposed assays. When a protein preservative solution is used for experiment termination, good results were obtained for subsequent 2-D gel electrophoresis. RNA was obtained when samples were preserved with RNAprotect (QUIAGEN). The use of either fresh-washed cell suspensions or water-adapted cultures as the inoculum provides flexibility for the flight experiment depending on the constraints of the mission. Short-term (1-2 days) of ambient temperatures during descent are not expected to have serious effects on the science return.
Bibliography Type:	Description: (Last Updated: 05/02/2017)
Abstracts for Journals and Proceedings	Pulcini, E.; Broadaway, S.; Pyle, B. "Pseudomonas aeruginosa virulence and proteomics in simulated weightlessness" American Society for Microbiology Annual Meeting Abstract I-033 , May-2004
Abstracts for Journals and Proceedings	Guadarrama, S.; Pulcini, E.deL.; Broadaway, S.C.; Pyle, B.H. "Pseudomonas aeruginosa growth and Exotoxin A production in static and Modeled Microgravity environments." American Society for Gravitational and Space Biology Abstract #55 , Nov-2004
Abstracts for Journals and Proceedings	Pyle, B.H. "Microgravity and Microbes: Ground and Spaceflight Experiments" New Zealand Microbiological Society Abstract , Nov-2004